

Sam Thiel

Mathematician and Musician. sthiel@oberlin.edu. <https://stolenquotient2.github.io/>

EDUCATION:

Oberlin College, Oberlin, OH. Diploma expected June 2026.

Declared Majors in Mathematics and Musical Studies. Honors Candidate in Mathematics.

Deerfield Academy, Deerfield, MA. Diploma received May 2022.

RELEVANT COURSEWORK (Mathematics):

- Complex Analysis (MATH 356)
- Real Analysis (MATH 358; Measure Theory and Functional Analysis)
- Harmonic Analysis (MATH 357)
- Foundations of Analysis (MATH 301)
- Probability (MATH 355)
- Topology (MATH 353)
- Group Theory (MATH 327)
- Computational Algebra (MATH 328)
- Linear Algebra (MATH 232)
- Mathematical Approaches to Music Theory (MUTH 347)
- Introduction to Computer Science (CSCI 150)

RESEARCH EXPERIENCE:

Maximal Functions and Maximal Inequalities, Oberlin Honors Project in Mathematics (with Chris Marx) (Sept. 2025–Present)

- Exploring the topic of Maximal Functions and Maximal Inequalities, and their applications to results in many subfields of analysis, including Harmonic Analysis, Measure Theory, Ergodic Theory, and Probability.
- The Honors Project will culminate in a paper on the subject, a technical presentation to a small selection of readers, as well as a more-accessible public talk, at the end of Spring Semester 2026.

Partial Results of the HRT Conjecture, Research Experience for Undergraduates, UWashingon-Seattle (June 2025–July 2025)

- Read and presented on the published partial results of the HRT Conjecture with a small group.

PROFESSIONAL EXPERIENCE:

Oberlin College and Conservatory, Oberlin, OH.

Research Assistant (with Chris Marx) (Aug. 2024–Present)

- Met weekly with Prof. Marx to work on his textbook on the intersection between Harmonic Analysis and Music.
- Composed in-text figures to help students visualize novel tools and results.
- Developed online resources to enhance student learning by providing hands-on examples.
- Designed end-of-chapter problems to provide essential practice in the textbook material.
- Proofread sections of the textbook for errors and clarity.

Teaching Assistant (For Susan Colley, Alexander Wilson) (Fall Semester 2024–Spring Semester 2025)

- Ran weekly 90-minute problem sessions each week for students taking **Group Theory** (MATH327).
- Collected, graded, and provided feedback on weekly homework assignments.
- Analyzed common mistakes and difficulties and maintained communication with the professor about these issues.
- Developed individual rapport with students in order to build a safe and supportive learning environment.

Homework Grader (For Susan Colley) (Fall Semester 2022–Spring Semester 2024)

- Collected and graded weekly homework for students taking **Linear Algebra** (MATH 232, proof-based) and **Multivariable Calculus** (MATH 231, non-proof based).
- Provided insightful comments and feedback to promote student learning.
- Analyzed common mistakes among students and relayed information back to the professor.

Presentations:

- **Costello Divisibility: Explorations of a Comedic Division Algorithm**, Talk, *Nebraska Conference for Undergraduate Wisdom in Mathematics 2026*, (to be given) Jan. 2026
- **Costello Divisibility: Explorations of a Comedic Division Algorithm**, Poster, *2024-2025 Oberlin Undergraduate Research Symposium*, Apr. 2025.

Publications:

- **Costello Divisibility: Explorations of a Comedic Division Algorithm** (with Alexander N. Wilson). *The College Mathematics Journal*, submitted.

Skills & Software:

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| • Microsoft Office Suite | • Python | • Ableton Live |
| • Google Suite | • GitHub | • MuseScore |
| • LaTeX/TexShop | • Affinity Designer | • Logic Pro |